BT-204 (GS)

B.Tech., I & II Semester Examination, November 2022

Grading System (GS)

BASIC CIVIL ENGINEERING & MECHANICS

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) Write about physical and chemical properties of concrete. **7**

(b) Explain about the different types of stair case used in building construction with a neat sketch. **7**

2. (a) Explain the individual components of the Theodolite with a sketch. **7**

(b) Narrate about the classification of stone used in construction industry. **7**

3. (a) Explain in brief about Remote sensing and its application in construction. **7**

(b) State and derive the Lami's theorem and solve the given problem using the theorem. **7**

4. (a) Discuss about the Contour line, Contour Interval, Horizontal Interval and use of Contour Map. **7**

(b) A block of mass M = 10 kg is sitting on a surface inclined at angle θ = 45°. Given that the coefficient of static friction is μs = 0.5 between block and surface, what is the minimum force F necessary to prevent slipping? What is the maximum force F that can be exerted without causing the block to slip? **7**

5. (a) Illustrate the difference between the Height of Collimation and Rise and Fall method. **7**

(b) The following consecutive readings were taken with a levelling instrument at intervals of 20 m: 2.375, 1.730, 0.615, 3.450, 2.835, 2.070, 1.835, 0.985, 0.435, 1.630, 2.255 and 3.630 m. The instrument was shifted after the fourth and eighth reading. The last reading was taken on a BM of RL 110.200 m. Find the RLs of all the points using Rise and Fall method and satisfy the answer with arithmetic check. **7**

6. (a) Differentiate the difference between Moment of inertia and Product of inertia. **7**

(b) Calculate the Moment of Inertia about the X-axis. (Figure based — see original paper.) **7**

7. (a) Write about the difference between Method of Joint and Method of Section. **7**

(b) Draw the Shear force and Bending moment diagram for the simply supported beam carrying a uniformly distributed load of intensity W kN/m throughout a span of length L m. **7**

8. Explain about the following terms: **14**

(a) Centroid and Centre of Gravity (b) Profile Cross-sectioning (c) Plane table and its used instruments

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